

CHAPTER 1

Introduction

1.1 Motivation

Unmanned Aerial Vehicles (UAVs) have proved their great capability in transporting cable suspended loads in recent years. They offer the ability to reach places that are dangerous or remote and can transfer equipment that is otherwise difficult to move [1]. This can be of great utility during rescue operations, construction, humanitarian aid and logistics. While early efforts focused on single multirotors, they often face engineering challenges due to their limited payload capacity, endurance and ability to control the full load pose. To counter these challenges, collaborative teams of multirotors have been proposed.

Multirotors were mainly used because they are more agile than fixed-wings carriers and can hover. However, these solutions tackle only the limited payload capacity and the full pose control problems. They still suffer from reduced operational range due to their lower energy efficiency. On the other hand, fixed-wings platforms offer better energy efficiency [2] and thus longer operational range and higher cruising speeds. Hence, fixed-wings platforms are better suited for applications that require long range. This comes at the cost of a more complex control architecture because fixed-wings carriers cannot simply hover in place.

Multiple fixed-wings UAVs need to coordinate among themselves in order to allow the payload to stay stationary while they are loitering. This coordination is also needed in many other scenarios in the trajectory control of the payload. For this reason, most previous efforts were using multirotors. Recent works by [3],[4] and [5] showed that it is possible to calculate coordinated trajectories for the carriers online, such trajectories would respect the non-stopping condition while also maintaining load tracking. The solution however, was based on a centralized controller that calculated all the trajectories and then assigns each carrier its trajectory. Additionally, to calculate the trajectories online it had to solve an NLP each time step. This makes onboard deploying of the solution particularly challenging due to the high computational cost.

In this work, we propose a decentralized solution based on Imitation Learning (IL) and Multi-Agent Reinforcement learning (MARL). The proposed solution also allows for onboard deployment as the computational cost per step is greatly reduced to the inference time. Additionally,

its decentralized nature allows it to be deployed with great flexibility in many applications.

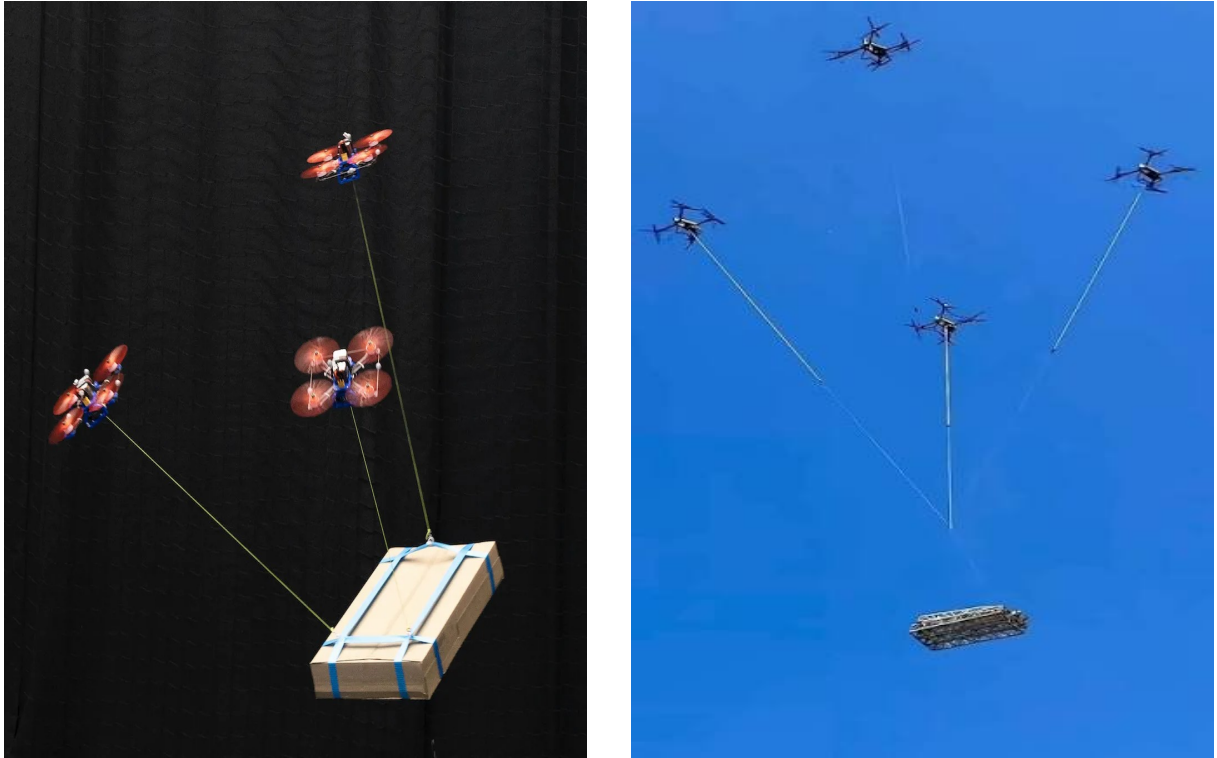


Figure 1.1: Cooperative teams of multirotor UAVs transporting a cable-suspended payload.

1.2 Main challenges

Fixed-wings UAVs present a unique and underexplored challenge in that they cannot hover. Since they always have to be moving, that means they will apply additional forces and torques on the payload other than the desired ones to move it along its trajectory. The main idea to solve this problem is to have these additional forces (referred to as internal forces) cancel each other out such that only the desired wrench (forces and torques) on the payload remains.

Hence, coordination between the carriers is a requirement. However, this coordination can still be achieved through well designed decentralized mechanisms depending on the load state and other local observations, without the explicit need for inter-carrier communication or measurements.

Fixed-wings UAVs also present a subtle but nontrivial problem, since the carrier needs to be always moving, the load state to carrier action map is not a well-defined function. In other words, we can have the same load state but different carrier actions in different times. Consequently, the carrier needs to take decisions based on memory or some understanding of time. Moreover, the actions of the carriers are not identical at the same moment in time and at the same payload state. This is due to the fact that the internal forces that will cancel each other need to be produced by different carrier trajectories.

Furthermore, since the policy is decentralized, each carrier will have to make its own ob-

servation of the load. We can imagine using an AprilTag on the load and each carrier has its own camera that sees it. Hence, each observation is going to be subject to its own noise and delay, leading to desynchronization in the load state between the different carriers. Therefore, the policy needs to learn good actions with limited and potentially outdated information. This naturally leads to a Partially Observable Markov Decision Process (POMDP) problem formulation. Reinforcement Learning (RL) provides a natural framework for learning control policies in such uncertain and partially observable environments. Finally, the problem has many other aspects that are beyond the scope of this thesis (e.g. enforcing a minimum turning radius).

The goal of this thesis is to answer whether a fully decentralized policy (using a combination of learning-based and classical methods) is able to coordinate the carrier trajectories even when they do not fully agree on the state of the system. Instead of relying on a centralized controller that has access to the full and true state, can they still learn a reactive policy that is able to stabilize the load and maintain their coordinated trajectories? Taking into consideration the carriers have their own local heterogeneous asynchronous observations. Additionally, the policy must be lightweight enough to deploy onboard and execute in real time.